INCH-POUND

MIL-F-52556G
22 October 1991
SUPERSEDING
MIL-F-52556F
20 April 1982

MILITARY SPECIFICATION

FILTER-SEPARATOR, LIQUID FUEL:

FRAME MOUNTED, 100 GPM CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 100-gallon per minute (gpm) (379 liter) capacity filter-separator for use in removing undissolved water and solid contaminants from petroleum fuels.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-T-60	- Tape: Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4330
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MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-F-8901	- Filter-Separators, Liquid Fuel: and Filter-Coalescer Elements, Fluid Pressure: Inspection Requirements and Test Procedures for.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-F-52308	- Filter Element, Fluid Pressure.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-C-81740	- Coatings, Aluminum and Aluminum Alloys (Metallic Compound Decomposition).
MIL-C-81751	- Coating, Metallic-Ceramic.
MIL-C-83488	- Coating, Aluminum, Ion Vapor Deposited.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

TA13217E5350	- Filter-Separator, Liquid, Fuel: Frame Mounted, 100 GPM Capacity.

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(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FSQ, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of Zinc or Iron and Steel.
D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.
D1.2 - Structural Welding Code - Aluminum.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P. O. Box 351040, Miami, FL 33135.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION (NMFTA)

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM FREIGHT CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The filter-separator shall be in accordance with Top Assembly TA13217E5350 and as specified herein. The filter-separator shall consist essentially of a vertical aluminum vessel with removable cover, a manually operated air vent, inlet and outlet connections, filter elements, canisters, differential pressure gage, a water level sight gage, a ground rod, a ground cable, attaching hardware, and a manually operated water drain valve. The filter-separator shall be mounted within a tubular rectangular frame. The general requirements, operating, and environmental conditions shall be in accordance with MIL-F-8901. The filter-separator shall have a rated flow of 100 gpm and a maximum working pressure of 75 pounds per square inch (psi).

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the filter-separator. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and as shown on the applicable drawings.

3.3.1 Material deterioration prevention and control. The filter-separator shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the filter-separator may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

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3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the filter-separator may be newly fabricated from recovered materials to the maximum extent practicable, provided the filter-separator produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the filter-separator.

3.4 Environmental conditions.

3.4.1 Operating temperatures. The filter-separator shall permit rated flow in any ambient temperature in the range of +125 to -25 °F.

3.4.2 Storage temperatures. The filter-separator shall not be damaged by storage at ambient temperatures from +160 to -50 °F.

3.5 Human factors. The filter-separator shall conform to the applicable human factors engineering criteria of MIL-STD-1472. Special design emphasis shall be given, but not limited, to general requirements (4), control/display integration (5.1), visual displays (5.2), controls (5.4), labeling (5.5), design for maintainer (5.9), and hazards and safety (5.13) of MIL-STD-1472, as applicable.

3.6 Maintainability.

3.6.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts as specified (see 6.8 and 6.9), shall be adequate to perform the intended function of assisting or conducting maintenance operations on the filter-separator.

3.6.2 Maintenance ratio. The filter-separator shall have a maintenance ratio of no greater than 0.03.

3.7 Reliability. The specified Mean-Time-Between-Failure (MTBF), shall be 140 hours when the filter-separator is operated at rated capacity and tested as specified in 4.5.2.5.

3.8 Transportability. The filter-separator shall withstand the dynamic vibrational stresses encountered in transportation and service. The filter-separator shall operate as specified herein when tested as specified in 4.5.2.7.

3.9 Performance.

3.9.1 Vessel (tank). The filter-separator vessel shall be fabricated of aluminum alloy as specified on the drawings. The vessel shall withstand a hydrostatic pressure of 115 psi without leakage, permanent deformation, or other defects that could harmfully affect the performance and serviceability of the filter-separator.

3.9.2 Deck plate. The vessel deck plate shall be fabricated of aluminum alloy and of the thickness as specified on the drawings. The deck plate shall withstand a minimum of 75 psi without leakage, permanent deformation, or other defects that could harmfully affect the performance and serviceability of the filter-separator.

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3.9.3 Differential pressure. The maximum differential pressure shall not exceed that specified by MIL-F-8901, table III.

3.9.4 Permanent separator stage. The permanent separator stage shall be as specified in MIL-F-8901, table III. The direction of flow shall be inside-to-outside.

3.9.5 Water removal. The filter-separator shall remove water as specified in MIL-F-8901, table III from influent fuel containing up to 5 percent (by volume) water.

3.10 Media. The media shall consist of five filter elements conforming to MIL-F-52308, and five canisters as shown on the applicable drawings. The Government will furnish 10 filter elements NSN 4330-00-983-0998 for each filter-separator (see 6.4), five to be installed in the filter-separator and five as replacements.

3.11 Differential pressure dial gage. The differential pressure gage shall conform to the dimensions shown and the requirements specified on the applicable drawing. The gage shall provide direct reading of the differential pressure across the filter-separator without manipulation of valves or subtraction of readings and shall perform without structural failure or leakage. The accuracy of the gage shall be such that any error shall not exceed ±2 psi.

3.12 Adapter and sampling probe. When specified (see 6.2), an adapter and sampling probe shall be furnished with the filter-separator. The adapter and sampling probe shall conform to the dimensions shown and the requirements specified on the applicable drawings.

3.13 Hardware. Bolts, nuts, and washers shall be fabricated from materials as specified on the applicable drawings. All metal parts in contact with fuel must be free of zinc, cadmium, copper, and their alloys. When ferrous materials are specified, the bolts, nuts, and washers shall be zinc coated in accordance with ASTM B 633, finish type IV, service condition SC-4, or aluminum coated in accordance with MIL-C-83488, type II, class 2, or in accordance with MIL-C-81740, type II, class 1, or coated in accordance with MIL-C-81751, type II. The selection of any treatment shall be in conformance with MIL-STD-889. Threads shall conform to FED-STD-H28.

3.14 Cover gasket replacements. Four O-rings, as specified on the applicable drawing, shall be furnished with each filter-separator as replacement items.

3.15 Sight gage gasket replacement. One gasket as specified on the applicable drawing shall be furnished with each filter-separator as a replacement item.

3.16 Identification marking. The filter-separators shall be identified in accordance with MIL-STD-130. The identification and instruction plates and markings shall be in accordance with the applicable drawing.

3.17 Treatment and painting. The exterior of the filter-separator normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), the top coat color shall be camouflage green 383 conforming to MIL-C-46168 or MIL-C-53039.

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3.17.1 Chemical coating. The interior of the filter-separator vessel and all components fabricated from aluminum alloy, which are in direct contact with the petroleum fuels, shall be anodized in accordance with MIL-A-8625, type II, class 1 to a minimum thickness of 0.0004 inch for casting and 0.0007 inch for wrought, or coated in accordance with MIL-C-5541, class 1A.

3.18 Cleaning and drying after tests. After conclusion of all tests, and after removal of the canisters and elements, the filter-separators shall be wiped clean and air dried at room temperature to exhaust all traces of fuel and water.

3.19 Workmanship. The filter-separators shall be free from defects such as misaligned components, incomplete welds, cracks, burns, leaks, and other defects that could impair the operation and serviceability of the filter-separator. All parts shall be clean and free from dirt, sand, grease, oil, and metal chips. Nonfunctional sharp edges and projecting points, which might present a hazard to personnel, shall be avoided.

3.19.1 Welders and welding.

3.19.1.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the following codes:

Welding Qualification, of the ASME, Section IX.
AWS D1.1, Structural Welding Code, Section 5, Qualification.
AWS D1.2, Structural Welding Code - Aluminum.

3.19.1.2 Welding (aluminum). Welding performed in the fabrication of filter-separators shall be in accordance with ASW D1.2. Welds shall be as specified on the drawings. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an

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acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article filter-separator shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The filter-separator shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling and examination. Sampling for examination and test shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, tables I and IIa. Samples shall be examined for the major and minor defects specified in 4.5.1. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Tests.

4.4.2.1 Individual. Each filter-separator shall be tested as specified in 4.5.2.1.a. and b. Failure of either test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. Filter-separators shall be examined as specified herein for the following defects:

Major

101. Any part or component not in accordance with QAP requirements as shown on the drawings.
102. Materials not as specified (see 3.3).
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).

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105. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.3).
106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the filter-separator (see 3.3.4).
107. Vessel (tank) not as specified (see 3.9).
108. Deck plate not as specified (see 3.9.1).
109. Differential pressure dial gage not as specified (see 3.11).
110. Adapter and sampling probe not as specified (see 3.12).
111. Hardware such as bolts, nuts, washers, and screw threads not as specified (see 3.13).
112. Cover gasket replacements not as specified (see 3.14).
113. Sight gage gasket replacement not as specified (see 3.15).
114. Chemical coating not as specified (see 3.17.1).
115. Filter-separators not cleaned and air-dried after testing (see 3.18).
116. Workmanship not as specified (see 3.19).
117. Weldings and castings not as specified (see 3.19.1 thru 3.19.1.2).

Minor

201. Dimensions, other than those identified by QAP data, not as specified.
202. Identification marking incorrect, illegible, or missing (see 3.16).
203. Instruction plates not as specified (see 3.16).
204. Treatment and painting not as specified (see 3.17).

4.5.2 Tests.

TABLE I. Test schedule.

First Article	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5
		Performance:		
X	X	Hydrostatic pressure	4.5.2.1.a	3.9.1
X	-	Differential pressure	4.5.2.1.c	3.9.3
X	X	Separator stage	4.5.2.1.b	3.9.4
X	-	5% water removal	4.5.2.1.d	3.9.5
		Temperature:		
X	-	High	4.5.2.2.1	3.4.1, 3.4.2
X	-	Low	4.5.2.2.2	3.4.1, 3.4.2
X	-	Human factors engineering	4.5.2.3	3.5
		Maintainability:		
X	-	Maintenance support	4.5.2.4.1	3.6.1
X	-	Maintenance ratio	4.5.2.4.2	3.6.2
X	-	Reliability	4.5.2.5	3.7
X	-	Differential pressure dial gage	4.5.2.6	3.11
X	-	Transportability	4.5.2.7	3.8

4.5.2.1 Performance. The filter-separators shall be tested to the following performance requirements as specified in MIL-F-8901, table III:

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a. Hydrostatic pressure test.
b. Permanent separator stage (inside-to-outside design).
c. Differential pressure.
d. Water removal, 5 percent.

Nonconformance to 3.9 through 3.9.5 or to the applicable requirements of MIL-F-8901 shall constitute failure of this test.

4.5.2.2 Temperature.

4.5.2.2.1 High temperature. The filter-separator shall be tested as specified in MIL-STD-810, method 501, procedure I. The maximum test temperature for storage shall be +160 °F, and the maximum operating temperature shall be +125 °F. The operating period of procedure I shall be 1 hour or the operating time required to check for any component failure due to high temperatures.

4.5.2.2.2 Low temperature. The filter-separator shall be tested as specified in MIL-STD-810, method 502.1, procedure I. The minimum test temperature for storage shall be -50 °F (-46 °C), and the minimum operating temperature shall be -25 °F (-32 °C). The operating period of procedure I shall be 1 hour or the operating time required to check for any component failure due to low temperatures.

4.5.2.2.3 Failure criteria. Nonconformance to 3.4 or to the applicable requirements for MIL-STD-810 shall constitute failure of either temperature test.

4.5.2.3 Human factors engineering. The filter-separator shall be examined by a Government human factors engineer for conformance to 3.5 requirements.

4.5.2.4 Maintainability.

4.5.2.4.1 Maintenance support. Maintenance operations required during testing shall be accomplished by operating personnel in accordance with the requirements of 3.6.1. Failure of the support elements to maintain the filter-separator as specified shall constitute failure of this test.

4.5.2.4.2 Maintenance ratio. The maintenance ratio shall be computed during first article testing to determine compliance with 3.6.2. The maintenance ratio is defined as the total active maintenance man-hours (scheduled and unscheduled) required to the total operating time. Nonconformance to 3.6.2 shall constitute failure of this test.

4.5.2.5 Reliability. Using the MTBF specified in 3.7, the filter-separator(s) shall be tested as specified in 4.5.2.1 and 4.5.2.2, with "accept" and "reject" criteria in accordance with figure 1. At completion of above testing, continue testing in accordance with 4.5.2.1.d until "accept" or "reject" decision is reached. A failure is defined as any malfunction which causes or may cause:

a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Damage to the filter-separator by continued operation.
c. Safety hazards to personnel.

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Nonconformance to 3.7 shall constitute failure of this test. Dependent failures or malfunctions occurring as the result of improper maintenance procedures or operator error are excluded from consideration as chargeable failures. Note: Replacement of filter elements shall not be considered a failure. Relevant failures shall be determined by authorized Government representative.

4.5.2.6 Differential pressure dial gage. Subject the gage to a hydrostatic pressure equal to the maximum scale pressure, and hold at this pressure for a minimum of 30 minutes. Within 10 minutes after the pressure is released and without recalibration or adjustment, readings shall be taken at three intervals on the scale; the anticipated initial operating pressure, the pressure at which the elements are changed, and the maximum scale pressure. The gage shall be tapped sharply with the finger at approximately the center of the scale area before each reading. A dead weight tester or mercury manometer with air pressure or calibrated check gage shall be used in calibrating the gage. Testing of the gage shall be in a temperature of 68 ±10 °F. Nonconformance to 3.11 shall constitute failure of this test.

4.5.2.7 Transportability. The filter-separator, shall be transported by a 2-1/2 ton cargo truck over cross-country at an average speed of 15 mph for a total of 100 miles. Damage to the filter-separator or components or nonconformance to 3.8 shall constitute failure of this test.

4.6 Inspection comparison. The Government may select filter-separators at any time during the contract production period and subject these filter-separators to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on filter-separators selected at random from those which have been accepted by the Government and will not include the previously inspected first article filter-separators. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison filter-separator to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison filter-separator(s) and shall be cause for refusal by the Government to continue acceptance of production filter-separators until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on filter-separators previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all filter-separators accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.7 Inspection of packaging.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, tables I and IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

118.	Canisters and filter elements not installed (see 5.1).
119.	Dust plug and dust cap not installed (see 5.1).
120.	Drain valve not open (see 5.1).
121.	Extra elements not preserved by one of the methods as specified (see 5.1.1).
122.	Extra O-rings and sight gage gasket not preserved as specified (see 5.1.2).
123.	Adapter and sampling probe not placed, cushioned, and secured within the box as specified (see 5.1.3).
124.	Consolidation of components not as specified (see 5.1.4).
125.	Each filter separator and components not packed into a shipping box and not secured within the box in a manner to prevent free movement and damage as specified for level A or B (see 5.2.1 and 5.2.2).
126.	Level C packing not in accordance with the referenced document (see 5.2.3).
127.	Marking missing, illegible, incorrect, or incomplete for level A, B, or C (see 5.3).

5. PACKAGING

5.1 Preservation. The filter-separator with canisters and filter elements installed shall not require preservation (see 6.7). The water drain valve shall be opened. Dust plug and cap shall be installed.

5.1.1 Extra elements. Each extra filter element shall be preserved by one of the following methods:

a. Each filter element shall be preserved in accordance with MIL-P-116, method IA-13.
b. Each filter element shall be inserted in a polyethylene bag fabricated from material conforming to L-P-378, type I, class 1, grade and finish optional, thickness not less than 0.0040, and closed by heat sealing. The bag containing the element shall be placed in a close-fitting box conforming to PPP-B-636, W6c. The box shall be closed and sealed as specified for method V in the appendix to the box specification using tape conforming to PPP-T-60, type IV.

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5.1.2 Extra O-rings and extra sight gage gasket. Each extra O-ring and the extra sight gage gasket shall be laid flat between two slightly larger pieces of fiberboard and each individually preserved in accordance with MIL-P-116, method IC-1.

5.1.3 Adapter and sampling probe. The adapter and sampling probe, when furnished, shall be placed in a close fitting box conforming to PPP-B-636, W6c, style optional. The contents shall be cushioned and secured within the box to prevent movement or damage.

5.1.4 Consolidation. The extra filter elements, O-rings, sight gage gasket and the adapter and sampling probe, when furnished, shall be consolidated together in one or more boxes as applicable, conforming to PPP-B-636, W5c, style optional. The contents shall be blocked, braced or cushioned as applicable to prevent movement or damage. Box closure and sealing shall be as specified for method V in accordance with the appendix to the box specification.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each filter-separator and the boxes containing the extra O-rings sight gage gasket, adapter and sampling probe and the extra filter elements required shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional. The contents shall be secured within the box in a manner to prevent free movement and damage. The box shall be closed in accordance with the appendix to the box specification. Strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.2.2 Level B. Each filter-separator and the boxes containing the extra O-rings, extra sight gage gasket, adapter and sampling probe and the extra filter elements required shall be packed together as specified for level A (see 5.2.1), except that the box shall be domestic type.

5.2.3 Level C. Each filter-separator and the consolidated components shall be packed together to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification or National Motor Freight Classification rules.

5.3 Marking. In addition to any special markings specified in the contract or purchase order, marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The filter-separator is intended for use in air field refueling systems, motor fuel servicing equipment, and military pipeline systems, for the removal of undissolved water and solid contaminants from aviation, diesel, or motor fuels.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of first article (see 3.2).
d. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
e. Finish paint, when other than as specified (see 3.17).
f. When the adapter and sampling probe is required and number to be procured (see 3.12).
g. Level of packing required (see 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.10.

6.5 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.6 Definition of QAP. A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.7 Levels of preservation. Levels of preservation have not been included, since the protection specified is the minimum acceptable. For purposes of preservation level marking, preservation shall be designated level A.

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6.8 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.9 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of filter-separators.

6.10 Subject term (key word) listing.

Filter-separator, air field refueling systems
Filter-separator, aviation fuel
Filter-separator, diesel fuel
Filter-separator, military pipeline systems
Filter-separator, motor fuel
Filter-separator, motor fuel servicing equipment

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - AS
Air Force - 99

Review activity:
Air Force - 84

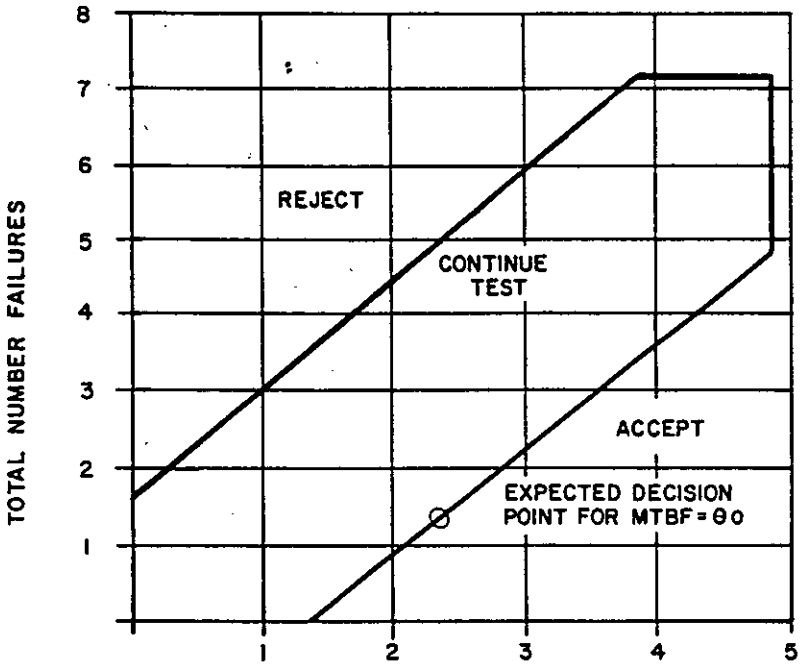
User activities:
Army - AT, AV
NAVY - MC

Preparing activity:
Army - ME

Project 4330-0090

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DECISION RISKS 20 %
DISCRIMINATION RATIO 2.0:1



NO. OF FAILURES	TOTAL TEST TIME * REJECT (EQUAL OR LESS)	ACCEPT EQUAL OR MORE
0	N/A	1.40
1	N/A	2.09
2	.35	2.79
3	1.04	3.48
4	1.73	4.17
5	2.43	4.87
6	3.12	4.87
7	3.81	4.87
8	4.87	N/A

* TOTAL TEST TIME IS TOTAL UNIT HOURS OF "EQUIPMENT ON" TIME AND IS EXPRESSED IN MULTIPLES OF THE SPECIFIED MTBF.

FIGURE 1. Accept-reject criteria.

X-3499

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-F-52556G	2. DOCUMENT DATE (YYMMDD) 911022

3. DOCUMENT TITLE
Filter-Separator, Liquid Fuel: Frame Mounted, 100 GPM Capacity

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Betty Taylor	b. TELEPHONE (Include Area Code) (1) Commercial (703) 664-1866 (2) AUTOVON 354-1866
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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